

Signal-Support Simulation Design

Product-Mixture MAP versus Viroli-Style Probit Gibbs

1 Purpose

The simulation asks when the proposed product-mixture MAP estimator can recover a high-dimensional binary factor mixture model. The working hypothesis is that performance is governed less by p alone and more by the strength and support of the probit signal: how large the nonzero item loadings are, how many items load on each factor, and how much signed cross-loading connects factors.

2 Data-Generating Model

Each replication generates

$$X_{ij} = \mathbf{1}\{Z_{ij} > 0\}, \quad Z_{ij} = \alpha_j + \boldsymbol{\lambda}_j^\top \mathbf{F}_i + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, 1).$$

The factor coordinates are independent marginal mixtures:

$$F_{ih} \sim \sum_{g=1}^{G_h} \pi_{hg} N(\mu_{hg}, \sigma_{hg}^2), \quad h = 1, \dots, H.$$

The generated factors are standardized to the canonical marginal mean-zero, unit-variance scale. Item intercepts are included in all simulations using the IFEval-like intercept generator.

3 Simulation Grid

Quantity	Values
Sample size	$n \in \{100, 200\}$
Items	$p \in \{500, 1000, 1500, 2000\}$ for Product MAP
Gibbs item counts	$p \in \{500, 1000\}$ only
Factors	$H \in \{5, 10, 15, 20\}$
Mixture components	$G_h = 2$ for all h or $G_h = 3$ for all h
Replications	25 per setting
Mixture separation	1
Block sizes	balanced or IFEval-like unbalanced
Loading design	Cross/IFEval-like sparse signed cross-loadings
Loading strength	weak: $ \lambda \sim U(1.25, 1.75)$; strong: $ \lambda \sim U(2.50, 3.00)$
Cross-loading probability	0.075 or 0.20
Cross-loading signs	random signs

The loading strength range is applied to both primary loadings and nonzero cross-loadings. Primary loadings are positive. Cross-loadings are randomly positive or negative. Representative loading matrices are generated by `scripts/sample_size/plot_dgp_loading_heatmaps.R`.

4 Product-Mixture MAP

The proposed estimator has three stages.

Low-rank probit signal. First estimate the item intercepts and centered rank- H probit signal $B = F\Lambda^\top$ by maximizing

$$\ell(\alpha, B) = \sum_{i,j} [X_{ij} \log \Phi(\alpha_j + B_{ij}) + (1 - X_{ij}) \log \{1 - \Phi(\alpha_j + B_{ij})\}], \quad \text{rank}(B) \leq H.$$

The implementation uses an EM-SVD update: compute conditional expectations of the latent probit variables, update intercepts, and project the centered working matrix onto rank H . This stage estimates the signal subspace directly and avoids repeated sampled- Z augmentation in the proposed method.

Rotation. The SVD of $\hat{B}$ gives left singular-vector scores spanning the signal subspace, but the axes are rotationally unidentified. The rotation stage searches for an orthogonal rotation whose coordinates are well fit by independent univariate Gaussian mixtures while also encouraging sparse loadings. The default implementation uses Riemannian rotation with an ℓ_1 loading penalty.

MAP refinement. Starting from the rotated scores, the refinement stage alternates updates of factor scores, item intercepts/loadings, and marginal mixture parameters. The loadings are estimated with an ℓ_1 penalty. Refinement uses monotone objective monitoring and stops only after the posterior objective is stable and, by default, all marginal mixture updates have converged.

The default Product MAP penalties are

$$\lambda_{\text{pretrain}} = \lambda_{\text{rotation}} = \lambda_{\text{refine}} = 10.$$

5 Viroli-Style Gibbs Baselines

Both Gibbs baselines fit the independent marginal mixture factor model with probit augmentation. Each iteration samples truncated latent probit variables, factor scores, marginal component labels, mixture weights, mixture means, mixture variances, item intercepts, and item loadings. Posterior draws are normalized to the same factor scale used in the DGP.

The Laplace baseline uses the Bayesian lasso scale-mixture representation for the loading prior, with penalty 10. The diffuse Gaussian baseline replaces that prior with a weak Gaussian loading prior. Both are run for 2000 iterations, discarding the first 1000 as burn-in. Effective sample sizes are recorded for parameters when enabled.

6 Parallelization

Product MAP uses one simulation chunk at a time by default, with internal parallel workers for independent marginal mixture fits, itemwise loading regressions, and subject-wise factor-score updates. Gibbs is parallelized across independent task chunks, so separate replications/configurations run concurrently. This avoids the overhead that previously made within-chain parallel Gibbs unattractive in these R implementations.

7 Recorded Results

Each row records the full DGP settings, method settings, convergence diagnostics, elapsed seconds, and recovery metrics:

$$\text{RMSE}(F), \quad \text{RMSE}(\Lambda), \quad \text{RMSE}(\alpha), \quad \text{RMSE}(\mu), \quad \text{RMSE}(\sigma^2), \quad \text{RMSE}(\pi).$$

The product estimator also records first-stage signal diagnostics, including relative Frobenius error for $\hat{B}$ and the sine of the principal-angle operator-norm error for the estimated signal subspace. These diagnostics are intended to separate failures of binary signal estimation from failures of rotation or MAP refinement.

8 Replication

From the repository root:

```
Rscript scripts/sample_size/run_final_product_viroli_simulation.R.
```

Progress plots can be regenerated at any point with

```
Rscript scripts/sample_size/plot_signal_support_simulation_progress.R.
```

The full raw output is written to `results/full/signal_support_grid`; selected GitHub-facing plots and tables are written under `results/selected_plots` and `results/selected_tables`.